

University of Pittsburgh

Introduction to Operating Systems
CS/COE 1550

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(Some slides are from **Silberschatz, Galvin and Gagne ©2013)**

Announcements

- Upcoming deadlines
  - Homework 4 is due next Monday 2/14
  - Project 1 due on 2/18

Previous lecture …

- It is easy to make mistakes when using semaphores
- Solution: Mutex and Condition Variables

Muddiest Points

Problem of the Day

Readers & Writers

- Many processes that may read and/or write
- Only one writer allowed at any time
- Many readers allowed, but not while a process is writing
- Real-world Applications
  - Database queries
  - We have this problem in Project 1

Semaphore-based Solution

Shared variables

```
int nreaders;
Semaphore mutex(1), writing(1);
```

Reader process

```
…
mutex.down();
nreaders += 1;
if (nreaders == 1) // wait if
  writing.down();  // 1st reader
mutex.up();
// Read some stuff
mutex.down();
nreaders -= 1;
if (nreaders == 0)      // signal if
  writing.up();         // last reader
mutex.up();
…
```

Writer process

```
…
writing.down();
// Write some stuff
writing.up();
…
```

Solution Tracing

- enterRead

Reader process

```
…
mutex.down();
nreaders += 1;
if (nreaders == 1) // wait if
  writing.down();  // 1st reader
mutex.up();
// Read some stuff
mutex.down();
nreaders -= 1;
if (nreaders == 0)          // signal if
  writing.up();    // last reader
mutex.up();
…
```

Solution Tracing

- read

Reader process

```
…
mutex.down();
nreaders += 1;
if (nreaders == 1) // wait if
  writing.down();  // 1st reader
mutex.up();
// Read some stuff
mutex.down();
nreaders -= 1;
if (nreaders == 0)          // signal if
  writing.up();   // last reader
mutex.up();
…
```

Solution Tracing

- doneRead

Reader process

```
…
mutex.down();
nreaders += 1;
if (nreaders == 1) // wait if
  writing.down();  // 1st reader
mutex.up();
// Read some stuff
mutex.down();
nreaders -= 1;
if (nreaders == 0)          // signal if
  writing.up();   // last reader
mutex.up();
…
```

Writer Events

- enterWrite

```
Writer process
…
writing.down();
// Write some stuff
writing.up();
…
```

Writer Events

- write

Writer process

```
…
writing.down();
// Write some stuff
writing.up();
…
```

Writer Events

- doneWrite

```
Writer process
…
writing.down();
// Write some stuff
writing.up();
…
```

Sequence 1

- W0 enterWrite
- W0 write
- R0 enterRead
- R1 enterRead
- R2 enterRead
- W0 doneWrite
- R2 read
- W1 enterWrite
- R2 doneRead
- W1 write

Reader process

```
…
mutex.down();
nreaders += 1;
if (nreaders == 1) // wait if
  writing.down();  // 1st reader
mutex.up();
// Read some stuff
mutex.down();
nreaders -= 1;
if (nreaders == 0)      // signal if
  writing.up();         // last reader
mutex.up();
…
```

Writer process

```
…
writing.down();
// Write some stuff
writing.up();
…
```

Sequence 2

- R0 enterRead
- R0 read
- R1 enterRead
- R1 read
- W0 enterWrite
- R2 enterRead
- R2 read
- R2 doneRead
- R1 doneRead
- R0 doneRead
- W0 write
- W0 doneWrite

Reader process

```
…
mutex.down();
nreaders += 1;
if (nreaders == 1) // wait if
  writing.down();  // 1st reader
mutex.up();
// Read some stuff
mutex.down();
nreaders -= 1;
if (nreaders == 0)       // signal if
  writing.up();          // last reader
mutex.up();
…
```

Writer process

```
…
writing.down();
// Write some stuff
writing.up();
…
```

Solution using Mutex and Condition Variables